

WASP-135b

R-2426

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-135_260717 · created 2026-08-22T00:08:43+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461238.673 +/- 0.016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.159 +/- 0.05
Transit depth (Rp/Rs)^2	2.53 +/- 1.58 [%]
Orbital Inclination (inc)	81.56 +/- 2.91
Airmass coefficient 1 (a1)	38.71 +/- 27.39
Airmass coefficient 2 (a2)	-2.99 +/- 1.41
Scatter in the residuals of the lightcurve fit is	121.44 %
Best Comparison Star	#1 - [417, 183]
Optimal Aperture	1.71
Optimal Annulus	12.15
Transit Duration (day)	0.034 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.159 ± 0.05 vs 0.139 (deviation 0.24σ)
Duration fit vs published	0.034 d vs 0.0687 d (deviation 1.0σ)
Mid-time O–C	-40.9 min
Depth SNR	1.9
Residual scatter	121.44 %

Light curve

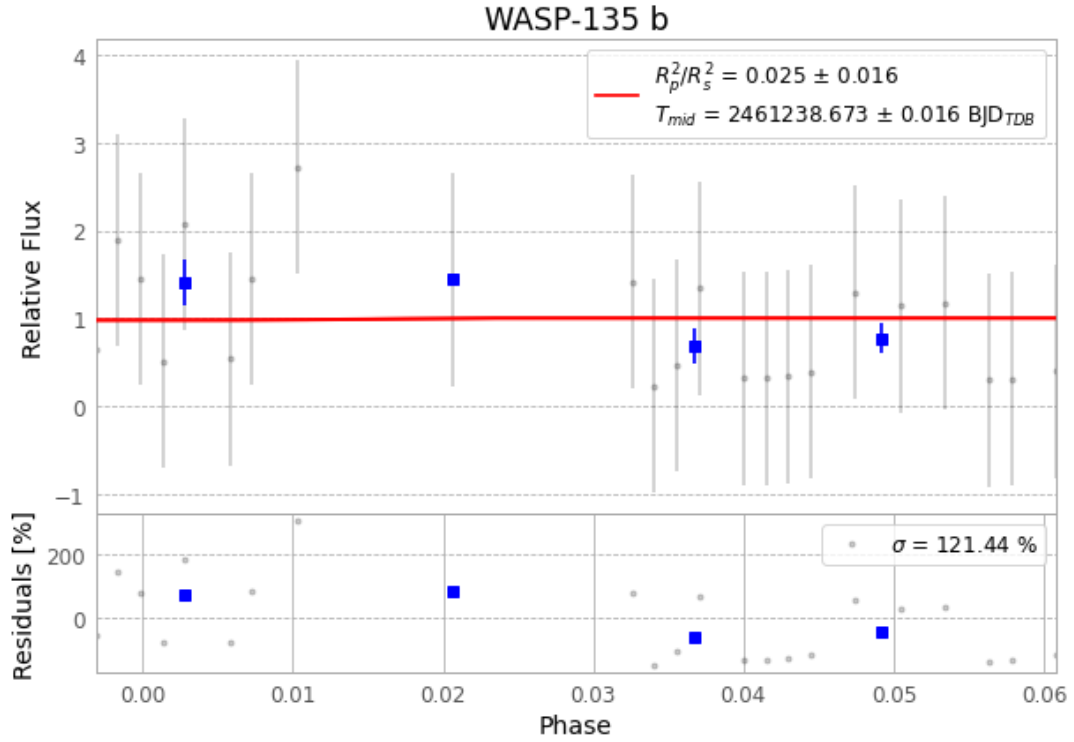


Figure 1 — FinalLightCurve_WASP-135 b_2026-07-16.png (SHA-256 b5bb0f0aad6b0b7d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [265, 220], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 73109a754bf899ad...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

46 FITS frames (30 MB), WASP-135260717035716.FITS → WASP-135260717061216.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-135-260717.log (b40f66e3be10...), FinalLightCurve_WASP-135 b_2026-07-16.png (b5bb0f0aad6b...), FinalLightCurve_WASP-135 b_2026-07-16.csv (825609fbcd62...)

Compute resources

Wall time 117.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-22 00:08Z.